

In Class Assignment Part I - Partial Derivatives

Group Member Names :

A. Partial Derivatives on a Surface

Use dry erase markers to label the following on your clear plastic surface.

1. Determine an orientation for the surface. Which way is positive x , y and z ?
2. Label a point, C, where f_x is a small positive number.
3. Label a point, D, where f_y is a large negative number.
4. Label a point, E, where $f_x = 0$ and $f_y = 0$.
5. Label a point, F, where $f_x = 0$ and $f_y \neq 0$.
6. Label a point, G, where $f_{xx} > 0$ and $f_{yy} > 0$.
7. Label a saddle point, H.

B. Partial Derivatives on a Contour Map

Using **Figure 1** and **Figure 2** below, complete the following.

8. For which points (T,P,R,A,B) is f_x positive?
9. What is the sign of $f_{xx}(P)$? Justify your answer.
10. Which is larger, $f_x(P)$ or $f_x(R)$? Justify your answer.
11. Mark a new point S on **Figure 2** such that $f_y(S) = 0$

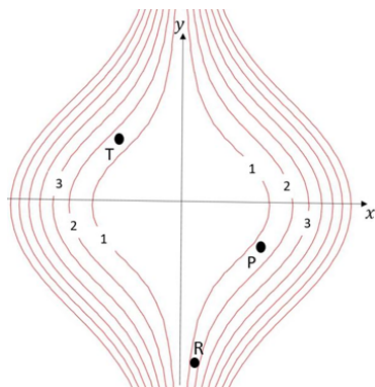


Figure 1

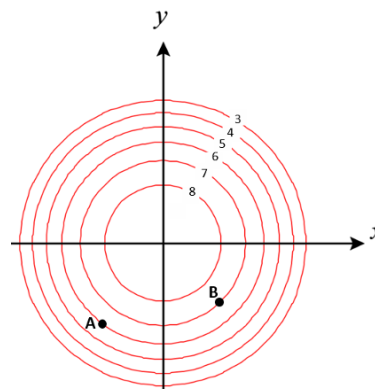
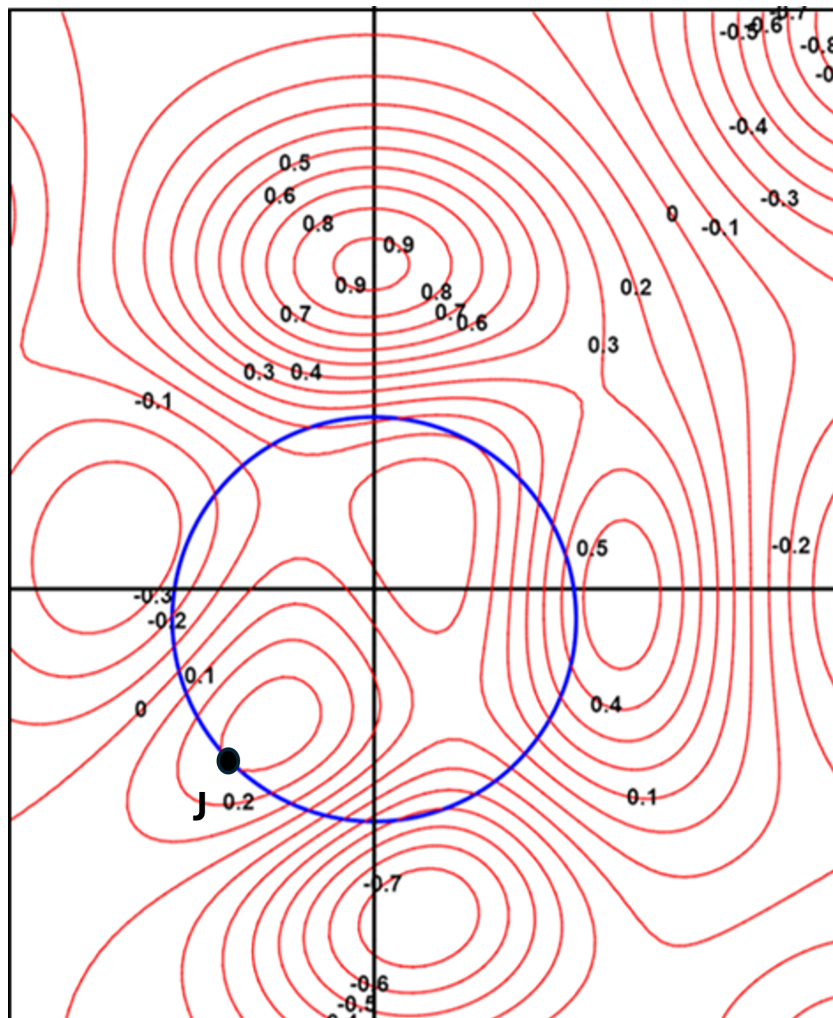


Figure 2

Show you instructor your work before beginning Part 2.

In Class Assignment Part II - Lagrange Multipliers

The below contour map shows the level curves of the function $z = f(x, y)$ which represents the elevation of a mountain range. The blue circle, $g(x, y) = c$, is a hiking trail. The trail reaches a maximum value at Point J.



12. Draw $\vec{\nabla}g(J)$ and $\vec{\nabla}f(J)$. (Estimate the magnitudes.)
13. Estimate and label the other 5 critical points on the hiking trail $g(x, y)$.
14. Draw the path of the hiking trail on your clear surface. *Hint : Align your clear surface to the contour map. Stand above the surface and look straight down while you trace the circle.*
15. Use the clear plastic surface to confirm the location of your 5 critical points. Label your points K,L,M,N,Q.
16. Draw estimated $\vec{\nabla}g$ and $\vec{\nabla}f$ for each of your 5 remaining points on the contour plot.
17. Go to the CalcPlot3d website, Explorations, Lagrange Multiplier Activity, Step 9. Adjust the blue slider on the bottom to confirm your $\vec{\nabla}g$ and $\vec{\nabla}f$.